

# Modeling Surface Vibrations and Their Role in Molecular Adsorption: A Generalized Langevin Approach

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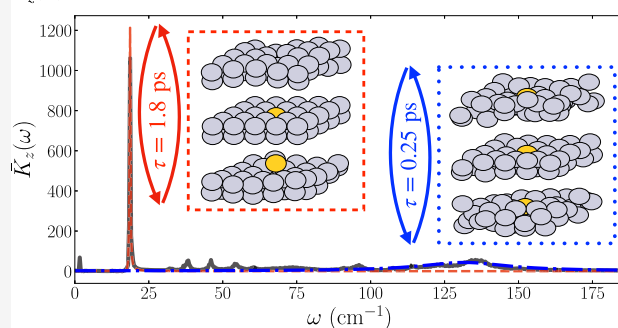


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**ABSTRACT:** The atomic vibrations of a solid surface can play a significant role in the reactions of surface-bound molecules, as well as their adsorption and desorption. Relevant phonon modes can involve the collective motion of atoms over a wide array of length scales. In this paper, we demonstrate how the generalized Langevin equation can be utilized to describe these collective motions weighted by their coupling to individual sites. Our approach builds upon the generalized Langevin oscillator (GLO) model originally developed by Tully. We extend the GLO by deriving parameters from atomistic simulation data. We apply this approach to study the memory kernel of a model platinum surface and demonstrate that the memory kernel has a bimodal form due to coupling to both low-energy acoustic modes and high-energy modes near the Debye frequency. The same bimodal form was observed across a wide variety of solids of different elemental compositions, surface structures, and solvation states. By studying how these dominant modes depend on the simulation size, we argue that the acoustic modes are frozen in the limit of macroscopic lattices. By simulating periodically replicated slabs of various sizes, we quantify the influence of phonon confinement effects in the memory kernel and their concomitant effect on simulated sticking coefficients.

 $\tilde{K}_z(\omega)$  = phonon frequencies weighted by coupling to surface atoms

## 1. INTRODUCTION

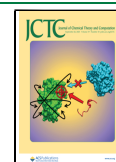
Heterogeneous catalysis, and surface chemistry more broadly, is an intricate dance between the electronic and nuclear degrees of freedom of the molecular reagents and underlying substrate. An aspect of this dance that is sometimes overlooked is the vibrational modes of the substrate surface atoms. Indeed, several recent studies have highlighted that these modes can significantly influence reactions at metal interfaces.<sup>1–4</sup> Methods that enable the study of these influences therefore have the potential to enhance our understanding of many surface chemical processes. One such method involves the generalized Langevin equation, which can reduce the complex dynamics of collective atomic surface motions to a single function: the memory kernel (or non-Markovian friction). The memory kernel is a time-domain description of the phonon density of states re-weighted by the coupling of each phonon mode to the surface atoms. In this paper, we demonstrate how atomistic data and simulations can be used to parameterize the memory kernel and derive insights from its functional form. Crucially, we show that the memory kernel has a bimodal form arising from the strong coupling of surface sites to both coherent acoustic oscillations and a manifold of modes near the Debye frequency. This form is modified by the maximum phonon wavelength set by the simulation's boundary conditions in ways that can significantly affect surface chemistry, as we illustrate by calculating sticking coefficients. Specifically, nanoscale boundary conditions, such as those of

atomistic simulation, cause the acoustic modes to oscillate on chemical timescales. We show that this effect can lead to a systematic decrease in sticking coefficients from the macroscopic limit. A GLE-based approach, such as we present here, can mitigate these unwanted finite-size effects without necessitating large and computationally expensive simulations.

Over the past four decades, a multiplicity of experimental and theoretical studies have demonstrated how surface vibrations modulate reaction dynamics at metal interfaces. Since at least the 1980s, it has been understood that laser-induced photoexcitation of a metal surface can activate surface desorption.<sup>5</sup> The explanation of this effect has largely been attributed to hot electrons dissipating energy into the phonon modes of the metal, which subsequently couple to molecular desorption.<sup>1,2,5–9</sup> Another interesting case study has been the dissociation of methane on a variety of FCC metal surfaces. Several papers have demonstrated that the response of the surface tunes the tunneling barrier for the dissociation of the carbon–hydrogen bond, leading to higher dissociation rates

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more consistent with experiment measurements than if one used a fixed surface.<sup>10–17</sup> Similar effects were observed for the dissociation of N<sub>2</sub> on Ru(0001); however, in that case, the reaction dynamics could still be well-described by treating the nuclei classically.<sup>3,18–21</sup> Separately, it has been noted that surface acoustic waves generated via piezoelectric drivers can, when applied at specific frequencies and polarizations, strongly activate otherwise unfavorable reactions (such as the oxidation of ethanol on Pd).<sup>4,22</sup> Inspired by both the wealth of work demonstrating the variety of ways surface vibrations influence chemical dynamics, we have revisited the GLE as a tool for understanding and simulating surface atom dynamics.

The generalized Langevin equation describes the non-Markovian stochastic dynamics of a particle in a condensed phase environment as

$$\dot{p}(t) = -\frac{dW}{dq}(t) - \int_0^t K(t - \tau)p(\tau) d\tau + R(t) \quad (1)$$

where  $q$  and  $p$  represent the particle's position and momentum, respectively, and  $W$ ,  $K$ , and  $R$  are functions that characterize the influence of the environment on particle dynamics. Specifically,  $W$  is the potential of mean force, a function of particle position that encodes all relevant spatial correlations between the environment and the particle,  $R$  represents the random forces on the particle, typically assumed to be time-correlated Gaussian noise, and  $K$  is the memory kernel, which characterizes the time-correlated influences from the environment.  $K$  is a time-dependent analogue of the Markovian friction constant and is related to  $R$  via the second fluctuation dissipation theorem (FDT)  $K(t) = \langle R(t)R(0) \rangle / (mk_B T)$ . As we elaborate upon throughout the rest of the paper,  $K$  is of central importance to determining the dynamics of the GLE.

Adelman and Doll<sup>23</sup> were the first to discuss in depth how the GLE could be used to model the effect of substrate phonons on a site within a purely harmonic lattice. Later, Tully developed the generalized Langevin oscillator (GLO) method, wherein the motion of a surface site is described via a GLE with a memory kernel that is given by a single exponentially damped sinusoid.<sup>24</sup> As we discuss in Section 2.3, such a memory kernel is equivalent to coupling the surface atom to a single dissipative (ghost) oscillator. Tully's GLO method has seen much success as a highly computationally efficient way of modeling surface dynamics, particularly in application to molecular beam scattering experiments.<sup>25–29</sup> However, modeling the dynamics of the surface with a single mode is a limiting approximation; in principle, surface atoms should couple to each normal mode of the lattice.

In this study, we extend Tully's GLO model to allow for a memory kernel of arbitrary complexity and examine when and how the properties of the memory kernel affect molecular adsorption and surface scattering. We call this model the lattice-generalized Langevin equation (LGLE). In order to parameterize the memory kernel, we use data taken from atomistic simulations. Crucially, in Section 4, we show that the qualitative properties of the memory kernel are independent of the atomistic model details, such as force field parameters.

The remainder of the paper is organized as follows. In Section 2, we present the formal theory behind the LGLE, focusing on methods to parameterize the equation and briefly reviewing the extended variable transformation used to map the non-Markovian dynamics to a bath of dissipative harmonic oscillators. In Section 3, we detail our simulation methods and

in Section 4, we analyze memory kernels derived from these simulations, varying the choice of force fields, elemental composition, solvation state, and lattice size. Finally, in Section 5, we discuss how the properties of the memory kernel generated via our approach affect adsorption, highlighting systematic errors that occur in sticking coefficients when basing the memory kernel on data from small periodic systems.

## 2. THEORETICAL BACKGROUND

The LGLE treats the motion of surface sites using eq 1, coarse graining over all other degrees of freedom in the solid/lattice. Interactions between surface sites and any molecular reagents are handled explicitly. The noise and memory terms in eq 1 arise from the influence of the bulk lattice on the surface atoms.

There are many approaches to parametrizing eq 1 for surface sites from atomistic information, and in this paper, we will discuss two in particular, henceforth termed the projection operator (PO) method and the correlation function (CF) method. The PO method is based on the approach detailed in the formative works of Adelman and Doll,<sup>23</sup> and Tully<sup>24</sup> and involves matrix operations applied to the mass-weighted Hessian. By contrast, the CF method requires the computation of time-CFs using MD (molecular dynamics) simulations and is commonly used in applications of the GLE to amorphous systems or liquid solutions.<sup>30–34</sup> Both approaches are useful and will be employed in subsequent sections.

**2.1. Projection Operator Method.** The PO method is a simple and easily interpretable way of deriving and parameterizing the LGLE. The method requires computing and diagonalizing the mass-weighted Hessian of the substrate lattice.

To begin, the many-body potential energy surface of the lattice  $U(q_1 \dots q_N)$  is expanded to second order about a local minimum

$$U(q_1 \dots q_N) \approx U_0 + \frac{1}{2} \sum_{ij} \frac{\partial^2 U}{\partial q_i \partial q_j} q_i q_j \quad (2)$$

where  $U_0$  is the potential energy at the minimum-energy configuration,  $\{q_i\}$  is the set of displacements of each atom from its equilibrium position, and  $N$  is the total number of atoms in the system. In this expansion, the force along any displacement is  $F_k = -\left(\frac{\partial U}{\partial q_k}\right) = -\sum_i \frac{\partial^2 U}{\partial q_i \partial q_k} q_i$ . By introducing mass-weighted coordinates  $x_i = \sqrt{m_i} q_i$  and the mass-weighted Hessian  $D_{ij}^2 = \frac{1}{\sqrt{m_i m_j}} \frac{\partial^2 U}{\partial q_i \partial q_j}$ , the resulting equation of motion may be written as

$$\ddot{\mathbf{x}} = -\mathbf{D}^2 \mathbf{x} \quad (3)$$

Two POs,  $\mathbf{P}$  and  $\mathbf{Q} = \mathbf{1} - \mathbf{P}$  are then used to separate this equation into system (surface site) and bath (remaining lattice) subspaces, respectively

$$\ddot{\mathbf{x}}_P = -\mathbf{D}_{PP}^2 \mathbf{x}_P - \mathbf{D}_{PQ}^2 \mathbf{x}_Q \quad (4)$$

$$\ddot{\mathbf{x}}_Q = -\mathbf{D}_{QP}^2 \mathbf{x}_P - \mathbf{D}_{QQ}^2 \mathbf{x}_Q \quad (5)$$

where  $\mathbf{x}_P = \mathbf{P}\mathbf{x}$  are the system degrees of freedom,  $\mathbf{x}_Q = \mathbf{Q}\mathbf{x}$  are the bath degrees of freedom and  $\mathbf{D}_{PP}^2 = \mathbf{P}\mathbf{D}^2\mathbf{P}$ , etc. In principle, these POs can take any form as long as they obey the properties of idempotency  $\mathbf{P}^2 = \mathbf{P}$  and orthogonality  $\mathbf{PQ} = \mathbf{0}$ .

However, if we wish  $\mathbf{x}_p$  to correspond to the displacement of a particular surface atom, it is most natural to choose  $\mathbf{P}$  to be a matrix with ones on the diagonal for the indices corresponding to the coordinate(s) of interest and zeros elsewhere

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \mathbf{D}^2 = \begin{pmatrix} \mathbf{D}_{PP}^2 & \mathbf{D}_{PQ}^2 \\ \mathbf{D}_{QP}^2 & \mathbf{D}_{QQ}^2 \end{pmatrix} \quad (6)$$

For simplicity, let us allow  $\mathbf{x}_p$  to be a scalar  $x_p$  corresponding to the displacement of a single surface atom along a single coordinate. Concomitantly, if we let  $N$  be the total number of sites in the lattice,  $\mathbf{D}_{PQ}^2$  becomes an  $1 \times (N - 1)$  row vector,  $\mathbf{D}_{QP}^2$  becomes an  $(N - 1) \times 1$  column vector, and  $\mathbf{D}_{PP}^2$  becomes a  $1 \times 1$  scalar and thus is henceforth unbolded.

We then proceed by solving eq 5 in terms of  $x_p$ , substituting that solution into eq 4, and subsequently transforming back from mass-weighted to standard coordinates. The final solution is of the form of eq 1 and describes the motion of a surface atom with

$$K(t) = \mathbf{D}_{PQ}^2 \mathbf{D}_{QQ}^{-2} \cos(\mathbf{D}_{QQ} t) \mathbf{D}_{QP}^2 \quad (7)$$

$$W(q) = m[-D_{PP}^2 + K(t = 0)]q^2 \quad (8)$$

$$\frac{R(t)}{\sqrt{m}} = -\mathbf{D}_{PQ}^2 [\cos(\mathbf{D}_{QQ} t) \mathbf{x}_Q(0) + \mathbf{D}_{QQ}^{-1} \sin(\mathbf{D}_{QQ} t) \dot{\mathbf{x}}_Q(0) + \mathbf{D}_{QQ}^{-2} \cos(\mathbf{D}_{QQ} t) \mathbf{D}_{QP}^2 \mathbf{x}_P(0)] \quad (9)$$

Disregarding some mathematical subtleties with respect to the third term in eq 9,<sup>35</sup>  $K(t)$  and  $R(t)$  above do satisfy the FDT. Note that because  $K(t)$  determines the properties of  $R(t)$  via the FDT and also the determines the deviation of  $W$  from a fixed lattice,  $K(t)$  is arguably the most fundamental quantity in eq 1, containing all the relevant information for how the bath modulates the system's dynamics. Throughout the rest of this paper, it will prove useful to analyze the Fourier transform of the memory kernel,  $\bar{K}(\omega)$ , which can be shown to be equivalent to the power spectrum of the noise  $R(t)$  by the Wiener–Khinchin theorem

$$\bar{K}(\omega) = \sum_i \frac{c_i^2}{\omega_i^2} \delta(\omega - \omega_i) \quad (10)$$

In the above equation,  $\omega_i$  are the normal mode frequencies of the bulk lattice Hessian  $\mathbf{D}_{QQ}^2$  and  $c_i$  are the coupling constants between the surface degree of freedom and  $i$ th normal mode,  $c_i = \sum_j [\mathbf{D}_{PQ}^2]_{ij} V_{ji}$ , where  $\mathbf{V}$  is a matrix with the eigenvectors of  $\mathbf{D}_{QQ}^2$  as columns. The index  $j$  runs over the rows of  $\mathbf{V}$  and  $i$  over columns. Note that we ignore the negative frequency components of the Fourier transform in eq 10 as they are simply a reflection of the positive components. Equation 10 reveals that the peaks of  $\bar{K}(\omega)$  are nothing more than the phonon frequencies of the lattice weighted by their relative coupling to the site of interest. We will use this powerful interpretation throughout the rest of our paper.

**2.2. Correlation Function Method.** A notable disadvantage of the PO method outlined in the previous section is that it relies on the assumption that the PES is harmonic (eq 3). Unfortunately, some systems exhibit significant anharmonicities, such as solvated surfaces or surfaces with defects. In cases such as these, the GLE parameters can be determined by

analyzing system CFs. By system, we are referring to the degrees of freedom of interest, which for our purposes correspond to the coordinates of the surface sites of the lattice. This method begins from recognizing that the random force  $R(t)$  must be uncorrelated with the system's momenta:  $\langle R(t)p(0) \rangle = 0$ . This identity can be considered prerequisite for  $R(t)$  to be properly interpreted as a “random” noise and is indeed consistent with eq 9.

By taking the equilibrium time CF of both sides of the generalized Langevin equation (eq 1) with the initial momentum, we find

$$\begin{aligned} \langle \dot{p}(t)p(0) \rangle + \left\langle \frac{dW}{dq}(t)p(0) \right\rangle \\ = - \int_0^t K(t - \tau) \langle p(\tau)p(0) \rangle d\tau \end{aligned} \quad (11)$$

Using MD simulation, the force–momentum CFs (left-hand side) and the momentum auto-CF may be computed, and subsequently eq 11 may be solved to find  $K(t)$ . Standard numerical methods, such as those based on the use of the fast Fourier-transform algorithm and convolution theorems lack the numerical accuracy to reliably solve eq 11.<sup>34,36</sup> The method we use to solve eq 11 is described in detail in the Supporting Information.

It should be noted that for a general, nonlinear, and/or time-dependent environment, the memory kernel is a nonlinear operator and the noise term may not be Gaussian.<sup>37</sup> In this paper, we assume that a linear GLE with Gaussian noise to be a sufficient model for the surface fluctuations of a solid lattice.

**2.3. Extended Variable Transformation.** Computing the integral over the memory kernel in eq 1 is computationally intensive, especially for systems with long correlation times. A common strategy for simplifying this calculation is to expand the GLE into a set of Markovian equations. These equations describe system bilinearly coupled with bath of dissipative, stochastic harmonic oscillators. The advantage of using a dissipative bath, as opposed to an energy conserving bath, is that an extended bath can often be represented with one or two dissipative oscillators and that doing so dramatically reduces the dimensionality of the equations of motion. Here, we briefly summarize the method, noting that excellent presentations of the same formalism can be found in refs 38 and 39.

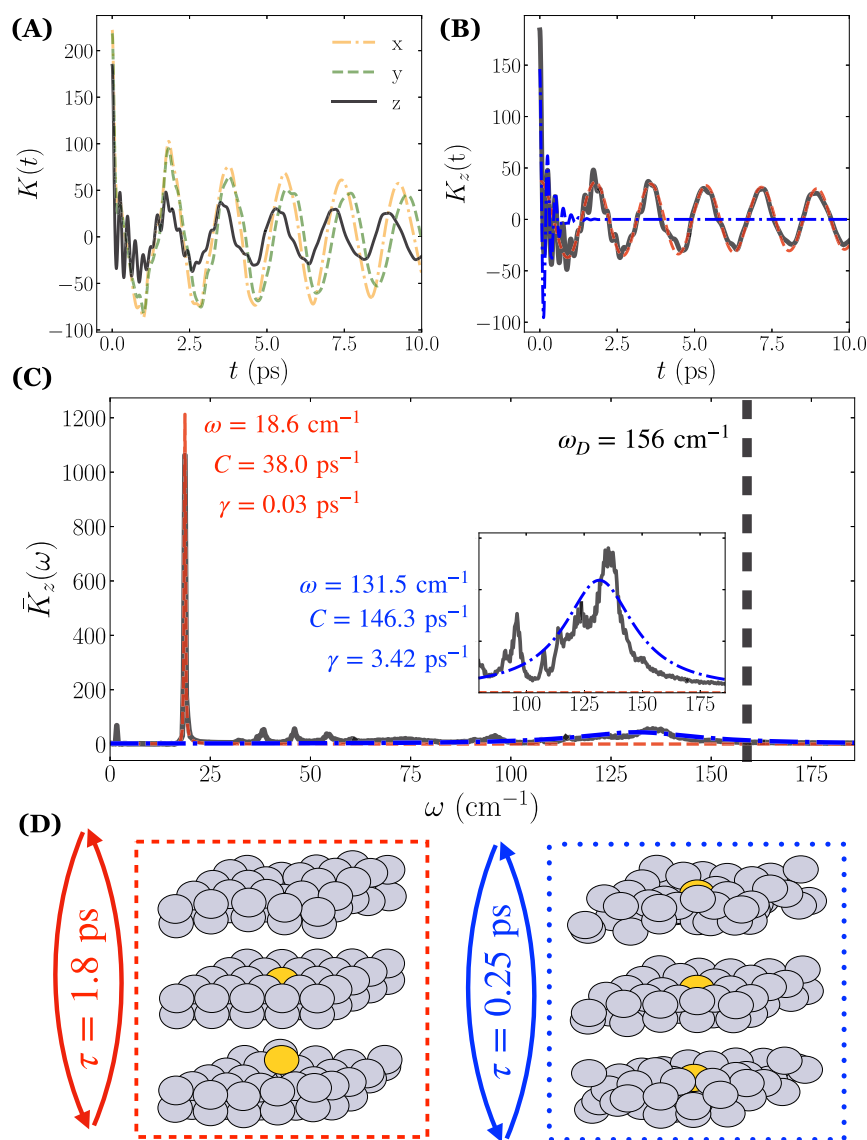
Given a GLE with a memory kernel that is a finite sum of exponentially damped sinusoids

$$K(t) = \sum_{i=1}^N e^{-\gamma_i t} (C_i \cos(\omega_i t) + D_i \sin(\omega_i t)) \quad (12)$$

the original non-Markovian equation of motion can be replaced with

$$\frac{d}{dt} \begin{pmatrix} p \\ \mathbf{b} \end{pmatrix} = \begin{pmatrix} -\frac{dW}{dq} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 & \mathbf{A}_{pb} \\ \mathbf{A}_{bp} & \mathbf{A}_b \end{pmatrix} \begin{pmatrix} p \\ \mathbf{b} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \mathbf{B}_b \end{pmatrix} (d\mathbf{W}) \quad (13)$$

Here,  $p$  is the system's momenta, and  $\mathbf{b}$  is a set of bath variables we must involve in time with our system.  $d\mathbf{W}$  is an array of uncorrelated Gaussian random variables satisfying  $\langle dW_i(t)dW_j(0) \rangle = \delta_{ij}\delta(t)$ , where  $\delta_{ij}$  is the Kronecker delta and



**Figure 1.** Memory kernel and random force power spectrum for surface sites of a Pt(111) lattice computed using an EMT force field. (A) Memory kernel for fluctuations in  $x/y$  (in-surface plane) and  $z$  (out of plane) directions. (B)  $z$  component of the memory kernel. Red and blue lines are two exponential sinusoids optimized to fit the computed memory kernel (grey line). (C) Power spectrum of  $z$  component of the memory kernel. The grey dashed vertical line corresponds to the experimental Debye frequency. (D) Depiction of lattice normal modes most associated with red and blue lines.  $\tau$  is the period of the respective normal mode. The yellow-highlighted sphere in the middle of the lattice represents the chosen adsorption site, which is fixed in space, while the surrounding grey atoms represent the fluctuating bulk lattice.

$\delta(t)$  the Dirac delta. The matrix  $\mathbf{A}_b$  is block diagonal with

entries

$$\mathbf{A}_b = \begin{pmatrix} 2\gamma_i & \sqrt{\gamma_i^2 + \omega_i^2} \\ -\sqrt{\gamma_i^2 + \omega_i^2} & 0 \end{pmatrix} \quad (14)$$

and  $\mathbf{A}_{pb}$  and  $\mathbf{A}_{bp}$  are arrays of form

$$\mathbf{A}_{pb} = \begin{pmatrix} \sqrt{\frac{C_i}{2} - 2\frac{D_i\omega_i^2}{\gamma_i}} & \sqrt{\frac{C_i}{2} + 2\frac{D_i\omega_i^2}{\gamma_i}} \\ \sqrt{\frac{C_i}{2} - 2\frac{D_i\omega_i^2}{\gamma_i}} & \sqrt{\frac{C_i}{2} + 2\frac{D_i\omega_i^2}{\gamma_i}} \end{pmatrix}, \quad \mathbf{A}_{bp} = \begin{pmatrix} \sqrt{\frac{C_i}{2} - 2\frac{D_i\omega_i^2}{\gamma_i}} \\ \sqrt{\frac{C_i}{2} + 2\frac{D_i\omega_i^2}{\gamma_i}} \end{pmatrix} \quad (15)$$

The matrix  $\mathbf{B}_b$  is related to  $\mathbf{A}_b$  by the equation

$$\mathbf{B}_b \mathbf{B}_b^T = k_B T (\mathbf{A}_Q + \mathbf{A}_Q^T) \quad (16)$$

which ensures that the ensuing dynamics obey the fluctuation–dissipation theorem.



Tully's GLO model is based on the same approach beginning from eq 12 and setting the number of terms  $N = 1$ . In the work we present in this manuscript, we determine the optimal values of  $N$ , as determined based on the analysis of  $K(t)$ . We drop the sin terms in eq 12, thus casting the memory kernel as a set of exponentially damped cosines and yielding a Lorentzian power spectrum of the form

$$\bar{K}(\omega) = \sum_{i=1}^N C_i \left( \frac{\gamma_i}{\gamma_i^2 + (\omega - \omega_i)^2} \right) \quad (17)$$

### 3. SIMULATION DETAILS

Simulations using Effective Medium Theory (EMT)<sup>40</sup> or Embedded Atom Method (EAM)<sup>41</sup> force fields for metal dynamics were performed using the Atomic Simulation Environment.<sup>42–47</sup> The parameters for these force fields were taken from refs 48 and 49, respectively. Simulations using Lennard–Jones force field, both solvated and in vacuum state, were performed using LAMMPS. Lennard–Jones force-field parameters were taken from ref 50. The solvent used was SPC/E.<sup>51</sup>

All simulations were performed in two steps. First, a temperature equilibration step was run for 50 ps at 300 K using a Langevin thermostat. Afterward, simulations were run in a constant energy ensemble using the velocity Verlet algorithm for 4 ns. Only data from the NVE step was used in subsequent analysis and calculations. All simulations were performed using periodic boundary conditions in the  $X/Y$  directions (parallel to the surface). Four atoms of the bottom row of the unit cell were rigidly constrained in order to remove center of mass motion.

For the surface scattering simulations used to generate data for Section 5, 5000 independent trajectories were averaged per value of the incident velocity to obtain sticking coefficients for GLE simulations, while 2000 independent trajectories were averaged for EMT simulations. Each trajectory was 20 ps in length, which we found to be adequate for the convergence of results.

### 4. MEMORY KERNELS AND POWER SPECTRA FOR SURFACE SITES IN METAL LATTICES

We begin by analyzing the memory kernel for the fluctuations of a single atom site in the surface of a  $4 \times 4 \times 4$  cell of Pt(111) with periodic boundary conditions. Results were calculated for each surface site individually and subsequently averaged. All memory kernels and power spectra presented in the main text are calculated via the CF approach (Section 2.2). We present results using the PO approach (Section 2.1) in the Supporting Information and will refer these results when necessary in the main text. Evidence for the convergence of the memory kernels presented here is given in Figures S1 and S2.

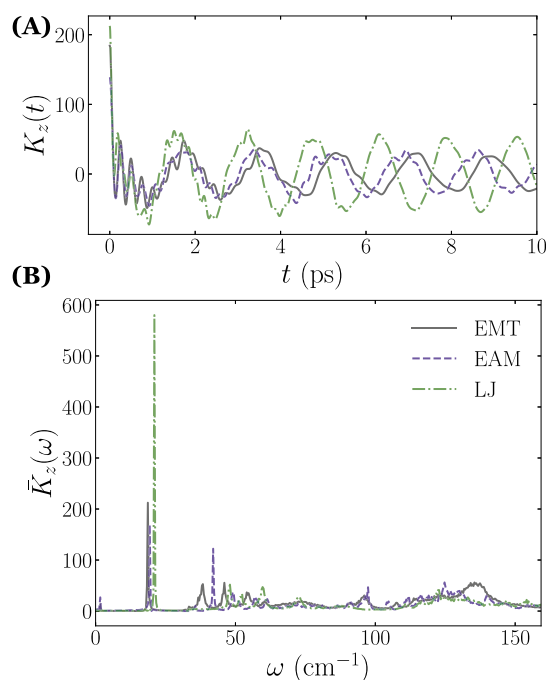
Figure 1A presents the orthogonal  $x$ ,  $y$ , and  $z$  components of the memory kernel calculated from all-atom simulations using an EMT force field. The  $x$  and  $y$  components arise from fluctuations in the plane of the lattice, and the  $z$  component arises from fluctuations normal to it. Note the anisotropy between the  $x$  and  $y$  components and the  $z$  component, a simple consequence of the difference in the number of the coordination number and anisotropy between longitudinal and transverse modes. In the analysis that follows, we focus only on  $K_z$ —the component of the memory kernel acting in the  $z$  coordinate—because fluctuations in this direction (perpendic-

ular to the metal surface) are most relevant to surface adsorption/desorption.

In Figure 1C, we present the noise power spectrum (Fourier transform of the memory kernel) of the  $z$  component  $K_z(\omega)$  specifically. As elaborated in Section 2, each peak in the power spectrum gives information about the lattice phonon modes and how they couple to the motion of a surface site. The value of  $\omega$  at each peak corresponds to the frequency of the mode, the width  $\gamma$  corresponds to the timescale of energy exchange or dissipation between the mode and the surface site, and the coefficient  $C$  indicates the coupling strength of the system to that particular mode.

The power spectrum in Figure 1C is essentially bimodal. The red peak—henceforth referred to as the acoustic peak—is centered at a low frequency ( $\omega = 18.6 \text{ cm}^{-1}$ ) and thus exchanges energy relatively slowly ( $\gamma = 0.03 \text{ ps}^{-1}$ ), while the blue peak—henceforth referred to as the Debye peak—is centered near the Debye frequency of Pt ( $\omega = 131 \text{ cm}^{-1}$ ) and exchanges energy relatively rapidly ( $\gamma = 3.42 \text{ ps}^{-1}$ ). By comparing the power spectrum computed via the CF method to the power spectrum computed via the PO method (Figure S4), it is possible to determine precisely which normal modes of the lattice are primarily responsible for these two peaks. These normal modes are illustrated in Figure 1D. The acoustic peak arises from longitudinal acoustic oscillations normal to the surface plane. Meanwhile, the Debye peak arises from many closely spaced normal modes near the Debye frequency, which consist of atomic-scale local oscillations.

In order to ensure the validity and transferability of our results, we tested the force fields other than EMT. These results are illustrated in Figure 2. The Lennard–Jones (LJ) model is based on a very different underlying physics than the



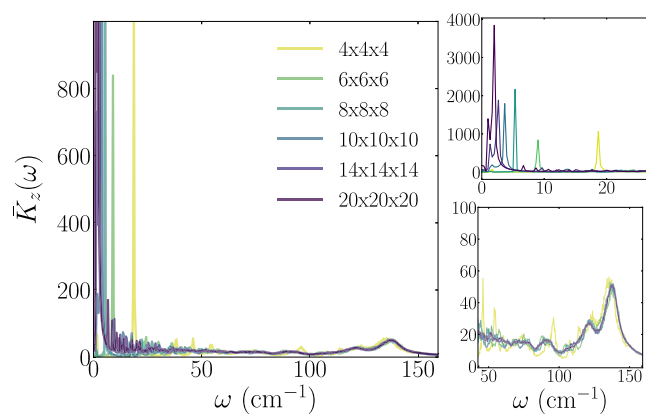
**Figure 2.** (A) Memory kernel and (B) power spectrum for surface site fluctuations of Pt(111) simulated using three different atomistic models: Effective Medium Theory, Embedded-Atom Method, and a Lennard–Jones model. The acoustic peak at  $20 \text{ cm}^{-1}$  of the power spectrum has been scaled by a factor of 0.3 for the purposes of visual clarity.

EMT/EAM models (LJ model uses only pairwise interactions, while EMT/EAM are both many-body potentials based on the local atom density). Despite this fact, all three models produce the same qualitative bimodal form. Much of the quantitative difference between the EMT/EAM and LJ models can be explained by the fact that the LJ model results in a stiffer lattice than EMT and EAM. The lattice stiffness can be roughly quantified in terms of the average value of the mass-weighted Hessian  $k_{\text{lat}} = \langle \mathbf{D}^2 \rangle$ . For EMT, the stiffness of the  $4 \times 4 \times 4$  Pt(111) lattice is 19.5 kJ/(mol nm<sup>2</sup>), for EAM it is 19.8 kJ/(mol nm<sup>2</sup>), and for LJ it is 37.1 kJ/(mol nm<sup>2</sup>).

We also tested lattices of different elemental composition and surface facets (Figure S6). Once again, although variations were observed in the location, widths, and heights of the primary peaks of the power spectrum, all of the lattices exhibited the same qualitative bimodal response. The apparent universality of this bimodal behavior might be considered a simple consequence of the structure of eq 10. Essentially, there is a trade-off between the amplitude of each mode, which decreases as the inverse square of the frequency and the density of modes, which increases sharply near the Debye frequency.

However, it is worth noting that this universality is not trivial. The bimodal behavior is not recovered in simple 1D systems with nearest-neighbor interaction (see refs 52 and 53 and Section S4 of the Supporting Information for more details) and therefore is an emergent property of the 3D metal lattice.

**4.1. Finite-Size Effects.** Periodic boundary conditions are used in MD simulations to mimic an infinite system but can still introduce finite size effects into phonon-mediated processes. Specifically, the size of the crystalline solid sets the maximum phonon wavelength. As such, increasing the size of the simulation should shift the frequencies of the acoustic modes, which in turn affects the shape of the memory kernel. In Figure 3, we confirm that the acoustic peak shifts to lower



**Figure 3.** Memory kernel and power spectra for surface site fluctuations of Pt(111) lattices of different sizes. The plots on the right zoom-in on the regions near to acoustic and Debye peaks, respectively.

frequency as the size of the lattice increases, while the Debye peak remains unchanged. Even when increasing the lattice size of  $\sim 45$  nm (8000 atoms), the memory kernel and power spectrum do not converge. In fact, the frequency ratio between the acoustic peak of different size lattices roughly agrees with the results of an isotropic wave equation, suggesting that the acoustic peak will never converge to a fixed frequency but

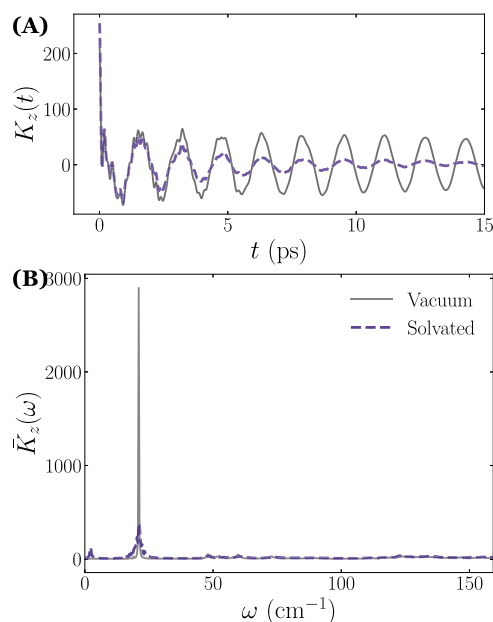
rather decrease like as  $1/L$  where  $L$  is the side-length of the lattice.

The demonstration that the memory kernel exhibits significant finite size effects has implications for the accuracy of simulation studies of surface phenomena. Perhaps most notably, this demonstration suggests that in the macroscopic limit, the frequencies of acoustic mode will be too low to affect any chemical dynamics at the surface. In other words, the acoustic mode is effectively frozen. Therefore, for observables that depend on memory, all finite-size simulations contain an intrinsic error which is purely kinetic in nature. We demonstrate this explicitly in Section 5.

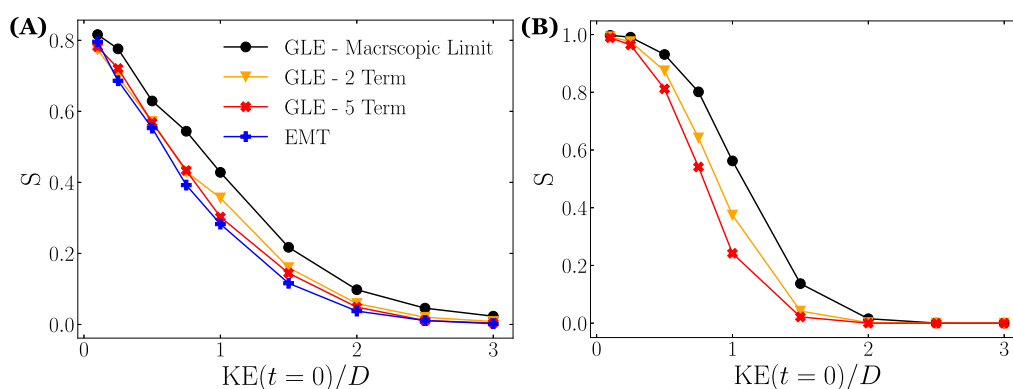
The size dependence of the memory kernel may also have ramifications for nanoparticle catalyst design because it demonstrates how nanoparticle vibrational modes behave quite differently than their macroscopic counterparts, much as their electronic modes do. Indeed, experimental studies of electron relaxation in metal supported nanoparticles have already shown that the phonon-mediated dissipation of electron energy depends strongly on the nanoparticle size.<sup>54</sup>

**4.2. Solvation Effects.** We now consider the effects of an adjacent solvent on the memory kernel of surface lattice vibrations. Thus far, we have assumed an idealized lattice in the limit of low pressure and substrate surface coverage. Most catalysts operate under conditions where there is significant surface coverage by solvent and/or reactive species. Here, we will explore how solvation affects surface site fluctuations by computing memory kernels for Pt(111) surfaces solvated in SPC/E water using the CF approach. We save the more difficult, yet still very important question, of surface coverage by physisorbed and chemisorbed species for future study.

Figure 4 demonstrates the difference between Pt(111) surfaces in vacuum versus in solvent. The primary difference is in the damping of the acoustic mode, whose coupling to the surface site motion is much smaller when the surface is solvated. This effect is likely attributable to the additional pressure exerted by the solvent, making large fluctuations in



**Figure 4.** (A) Memory kernel and (B) noise power spectra for surface site fluctuations of Pt(111) simulated using a Lennard–Jones model with and without SPC/E solvent.



**Figure 5.** Sticking probabilities  $S$  as a function of the ratio of the incident kinetic energy to the well depth  $KE(t=0)/D$ . (A) Results for Morse PES with  $D = 6.62$  eV. (B) Results for Morse PES with an increased well-depth,  $D = 30.62$  eV. The blue curve uses an all-atom simulation using an EMT force field to treat the metal degrees of freedom. The red and orange curves use the LGLE (eq 1) parameterized from a  $4 \times 4 \times 4$  EMT simulation to treat the metal. The red curve uses only two damped sinusoids to fit  $K(t)$ , while the orange curve uses a five-term fit to give a more accurate estimation of the memory kernel and power spectrum (see Figure S7.). The black curve corresponds to the extrapolated macroscopic limit of the LGLE, wherein the surface site motion is coupled to only to the Debye mode.

the direction normal to the surface plane more energetically costly. The damping of the acoustic mode suggests that the finite-size effects discussed previously are likely far less important for solvated surfaces than they are for surfaces in the gas phase.

## 5. HOW MEMORY AFFECTS ADSORPTION AND SCATTERING

Molecular beam scattering experiments are an invaluable tool for understanding the properties of surface reactions, elucidating information about binding potential energy surface and energy dissipation rates of the lattice.<sup>55–57</sup> When the incident particles are sampled from an appropriate thermal distribution, the surface sticking probability can be shown to be proportional to the adsorption rate constant. Tully's GLO model is often used in simulations of surface scattering (either reactive or non-reactive) as an efficient computational method for describing energy loss to the lattice during the scattering process.<sup>25–29</sup> In this section, we employ the LGLE for the same purpose, specifically studying the differences between the finite-size limit and the macroscopic limit (when the low-energy acoustic modes are held fixed).

We demonstrate our approach on the simple case of the non-reactive scattering of Argon on Pt(111) in the direction normal to the surface. The PES describing the interaction between the argon atom and the Pt surface is taken to be of Morse form

$$U(\Delta z) = D[1 - e^{-a(\Delta z - z_0)}]^2 \quad (18)$$

where  $\Delta z$  is the vertical distance between the Ar and the nearest Pt atom,  $D$  is a parameter which controls the depth of the PES well,  $a$  controls the width of the well, and  $z_0$  is the location of the well's minimum. The values of these parameters were fit from DFT calculations presented in ref 58 using a van der Waals density functional (vdW-DF2) and are presented in Table S1. Each scattering trajectory was initialized outside the Morse well at a distance of  $z = 15$  Angstroms, with a fixed initial velocity directed into the well and randomized lattice atom positions/velocities drawn from a thermal distribution at 300K. From these trajectories, the sticking probability,  $S$ , was calculated as the ratio of trajectories that remain trapped

within the well ( $z \leq 15$ ) Angstroms after a collision with the surface.

Figure 5A illustrates variations in the sticking probability for using four different simulation approaches described in the previous sections. The blue, red, and orange curves of Figure 5A largely agree with one another, illustrating that the LGLE accurately captures the dynamics of the force field it is parameterized from. More interesting however is the consistent increase in the sticking probability between the nanoscale lattices (either modeled with EMT or GLE) and the macroscopic limit. This discrepancy can be qualitatively explained by the relative dissipation rates of the acoustic and Debye modes. Since nanoscale lattices couple the motion of surface atoms to the acoustic modes and these acoustic modes dissipate energy much slower than the Debye mode, collisions with nanoscale lattices are more elastic. This effect can be observed more explicitly by studying histograms of the energy dissipated over many scattering trajectories (Figure S8).

In Figure 5B, we study the scattering probability of a stickier particle with a well-depth that is nearly 5 times greater. Increasing  $D$  increases the effective coupling between the adsorbate and the metal phonon bath, exacerbating the finite-size effects seen in Figure 5B. The results in Figure 5 highlight an error that is inherent to nanoscale simulations of surface–molecule interactions. This error is due purely to the phonon confinement effects placed by the boundary conditions and can be exacerbated by errors in the adsorbate's binding energy to the metal.

Finally, we also tested how the simulation approaches examined in this section affect desorption rate constants using the same Morse potential (eq 18). We did so because it is well-known that the rate constants do depend sensitively on the character of the memory kernel. The results are presented in the Supporting Information (Figure S9). Interestingly, no significant difference was found in the desorption rate constant between LGLE models parameterized from nanoscale simulation and the LGLE in the macroscopic limit. We believe this result indicates that more work is necessary to determine for what observables and under what conditions, the phonon-confinement effects discussed in this section are significant or not.



## 6. CONCLUSIONS

In this paper, we presented the LGLE, a model for simulating the effects of lattice phonons on surface atoms. The most important parameter in this model is the memory kernel. We parameterize the memory kernel using data from MD simulations, showing that it has a universal bimodal form due to coupling to both coherent acoustic oscillations as well as modes near the Debye frequency. This bimodal form is non-trivial, as it is not recovered in exactly solvable 1D systems with nearest-neighbor interactions. Since the frequency of the acoustic oscillations depends on the size of the lattice and nanoscale MD simulations impose unphysical phonon confinement effects, observables which depend on surface phonons will also contain artifacts. We showed that this was indeed the case for the surface trapping probability for a simple system of Argon on Pt(111).

The advantages of the LGLE model are multifold. The first advantage is, its computational efficiency, as it reduces the  $N$  degrees of freedom of the lattice, to only a small handful on terms needed to describe the motion of a surface site. This dimensionality reduction makes the LGLE much faster than all-atom force fields and particularly well-suited for the simulation of trajectory ensembles. The second advantage is the insight that can be gained from studying the memory kernel, as we illustrated throughout this paper. The third advantage is the transferability of the model. Once the LGLE is parameterized for a given type of lattice, any surface reaction with that lattice can use the same LGLE, given that the thermodynamic conditions (temperature/pressure/surface coverage/solvation) are roughly the same.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.3c00473>.

Analysis of convergence of memory kernels using CF method, a comparison of memory kernels using CF and PO methods, memory kernels for other metal lattices other than Pt(111), memory kernels for 1D harmonic chains, and further details on scattering/desorption simulations (PDF)

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## Notes

The authors declare no competing financial interest.

Data that support the findings of this study are available from the corresponding author upon reasonable request. The code used to process memory kernels and run GLE simulations is available on GitHub <https://github.com/afarahva/glepy>.

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